

Technology Stack

Date	2 July 2026
Team ID	(Your Team ID)
Project Name	India's Agriculture Crop Production Analysis
Maximum Marks	2 Marks

Define Your Technology Stack

Identify the programming languages, frameworks, databases, visualization tools, and machine learning libraries used to analyze India's agriculture crop production data. The selected technology stack enables efficient data processing, visualization, prediction, and report generation.

Technology Stack Details

S.No	Architecture Component/Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	Streamlit	Streamlit provides an interactive web interface for uploading datasets, selecting crops and states, visualizing trends, and displaying analysis results with minimal development effort.
2	Backend / Server-Side	Python	Python is used for data preprocessing, statistical analysis, crop production prediction, and integrating machine learning models. It offers extensive support for data science libraries.
3	Database / Data Storage	CSV Files / SQLite	Agricultural datasets are stored in CSV files or SQLite for efficient data management, retrieval, and analysis during application execution.
4	Data Analysis	Pandas & NumPy	Pandas is used for data cleaning and manipulation, while NumPy performs numerical computations required for agricultural data analysis.
5	Visualization	Matplotlib & Plotly	Used to generate interactive charts, crop production trends, state-wise comparisons, seasonal analysis, and yield distribution graphs.
6	Machine Learning	Scikit-learn	Implements regression and prediction models to estimate future crop production and identify production patterns.
7	Report Generation	ReportLab	Generates downloadable PDF reports containing crop statistics, visualizations, and analysis summaries.

This technology stack is well-suited for developing an **India's Agriculture Crop Production Analysis** system, providing reliable data processing, insightful visualizations, predictive analytics, and automated report generation.

S.NO	ARCHITECTURE COMPONENT/LAYER	TECHNOLOGY CHOSEN	JUSTIFICATION / PURPOSE
4	Cloud / Hosting / Deployment (e.g., AWS, Azure, Google Cloud)	Docker & AWS (EC2 / ECS / S3)	Docker ensures consistent deployment across all environments. AWS EC2 / ECS provides scalable and reliable compute resources for running the application. Amazon S3 is used for secure storage of large agricultural datasets, reports, and generated visualizations.
5	Version Control & CI/CD (e.g., GitHub, GitLab, Jenkins)	GitHub & GitHub Actions	GitHub enables efficient version control and collaboration among team members. GitHub Actions automates testing, code quality checks, and deployment pipelines for continuous integration and delivery.
6	Third-Party APIs / Other Tools (e.g., Weather API, Soil Data API, Google Maps API)	OpenWeather API, SoilGrids API & Google Maps API	Weather APIs provide real-time climate data for analysis. Soil data APIs offer information on soil properties. Google Maps API helps visualize geographical distribution of crop production across states and districts.
7	Data Sources (e.g., data.gov.in, ICAR, Survey Reports)	Government Open Data Portals & Agriculture Datasets	Public datasets from data.gov.in, ICAR, and Ministry of Agriculture are used for historical crop production data, area, yield, and other agricultural indicators.
8	Monitoring & Logging (e.g., AWS CloudWatch, Sentry)	AWS CloudWatch & Sentry	AWS CloudWatch monitors application performance and server health. Sentry helps track errors and exceptions in real time to ensure system reliability.